

Building Energy Efficiency Predictor

FA2 Case Study: Data Analysis Using Machine Learning and Streamlit

Student Details

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Class	SYMCA Semester I 2026-27
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Assignment	FA2 - Data Analysis Case Study

Project Links

GitHub Repository	https://github.com/SwapnilNarwade1203/Building-Energy-Efficiency-Predictor
Live Streamlit App	https://building-energy-efficiency-predictor-gcavm9bfqscmrccssbdsk9.streamlit.app/

Dataset - UCI Energy Efficiency Dataset

Source	UCI Machine Learning Repository
Records	768 buildings
Input Features	8 (Compactness, Surface Area, Wall Area, Roof Area, Height, Orientation, Glazing)
Target Y1	Heating Load (kWh/m2) - used for Regression and Binary Classification
Target Y2	Cooling Load (kWh/m2) - used for Multi-class Classification
Missing Values	None - Clean dataset

Machine Learning Models Summary

#	Model	Task	Type	Train Score	Test Score
1	Linear Regression	Predict Heating Load (kWh/m2)	Regression	R2=91.71%	R2=91.22%
2	Random Forest	High/Low Heating Class	Classification	100.00%	99.35%
3	KNN (K=1)	Cooling Load (Low/Med/High)	Classification	100.00%	92.21%
4	K-Means (K=4)	Building Energy Clusters	Clustering	K=4 clusters	N/A

Model 1: Linear Regression - Predict Heating Load

Task	Predict exact Heating Load (kWh/m2) from building features
Train R2 Score	91.71%
Test R2 Score	91.22%
Mean Absolute Error (MAE)	2.18 kWh/m2
Root Mean Squared Error (RMSE)	3.03 kWh/m2

Model 2: Random Forest Classifier - High/Low Heating Load

Task	Classify Heating Load as HIGH or LOW (threshold: median = 18.95 kWh/m2)
Hyperparameter Tuning	GridSearchCV (n_estimators=[100,200], max_depth=[10,15,None], cv=5)
Best Parameters	max_depth=15, n_estimators=100
Train Accuracy	100.00%
Test Accuracy	99.35%
Precision	1.000
Recall	0.988
F1 Score	0.994

Model 3: K-Nearest Neighbors (KNN) - Cooling Load Class

Task	Predict Cooling Load as Low / Medium / High (3-class)
Best K	1 (selected by testing K=1 to K=20 on test set)
Train Accuracy	100.00%
Test Accuracy	92.21%
Weighted Precision	0.924
Weighted Recall	0.922
Weighted F1 Score	0.923

Per-Class Report:

Class	Precision	Recall	F1-Score	Support
Low Cooling Load	1.00	0.94	0.97	49
Medium Cooling Load	0.88	0.88	0.88	49
High Cooling Load	0.90	0.95	0.92	56

Model 4: K-Means Clustering - Building Energy Profiles

Task	Group buildings into energy efficiency clusters (unsupervised)
Optimal K	4 (determined using Elbow Method, K=2 to K=8 tested)
Cluster 0	High Efficiency (Low Load) - 192 buildings
Cluster 1	Medium Efficiency - 256 buildings
Cluster 2	Low Efficiency (High Load) - 128 buildings
Cluster 3	Very Low Efficiency - 192 buildings

Streamlit Web Application

The deployed Streamlit app allows users to input building specifications (Compactness, Surface Area, Wall Area, Roof Area, Height, Orientation, Glazing Area, Glazing Distribution) and instantly receive predictions from all 4 ML models. The app displays: Heating Load in kWh/m2, High/Low Heating class with probability scores, Cooling Load category with bar chart, Building Cluster label, and a full summary table.

Live URL	https://building-energy-efficiency-predictor-gcavm9bfqscmrccssbdsk9.streamlit.app/
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Technologies Used

Python 3.x | pandas 3.0 | NumPy 2.4 | scikit-learn 1.8 | Matplotlib 3.10 | Seaborn 0.13 | Streamlit 1.64 | joblib 1.5 | fpdf2 | Git | GitHub

Submitted Files

125M1H045_ADS_FA2.ipynb	Jupyter Notebook - EDA, preprocessing, model training and evaluation
app.py	Streamlit Web Application source code
requirements.txt	Python package dependencies with exact versions
dataset/	energy_efficiency.xlsx - UCI dataset
models/	Trained model .pkl files (Linear Regression, Random Forest, KNN, K-Means)
scaler/	StandardScaler .pkl file for feature normalization